Penalising the Right Answer

How Fairness Audit Infrastructure Encodes Normative Commitments, and What Three Lending Audits Revealed

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ABSTRACT

A fairness audit is a claim about a decision that affects people. The infrastructure that produces such claims — metric libraries, reporting floors, scoring harnesses, quality checklists — is usually treated as neutral apparatus. It is not: it infers normative categories from syntactic features, and the inference fails hardest on the inputs where judgement was most needed. We demonstrate this four times over a lending audit, twice against instruments we built ourselves.

A subgroup reporting floor of $n \geq 15$, a standard and defensible statistical commitment, suppressed five of eleven cells in a mortgage audit. Three of them were worse than anything the audit reported, so it named the wrong group as most disadvantaged and understated severity by two bands: it returned MODERATE at 0.8169 while the worst real disparity was 0.1917. Auditing the full population instead of a test split makes those groups visible and moves the worst reported cell to 0.6734. A refusal detector flagged six correct answers as refusals because they fell below a character threshold under a prompt demanding terseness; applied to a second provider it mechanically zeroed fourteen cycles, moving that model's mean score from 70.0 to 19.79, and a permanent block was written into our configuration on that evidence. A fabricated-metric detector flagged a newer model five times as often as its predecessor — but 67% of those flags are values derivable from the provided context, mostly ratios restated as percentages, so the better-reasoning model is penalised for showing its work. And auditing age on the 40+ threshold borrowed from employment law returns near parity (0.9791) where the cut the credit statute specifies, 62+, returns the largest age disparity in the data (0.8897).

We also report what this cost us. Three previously published findings of ours do not survive: a CRITICAL intersectional ratio of 0.6087 that becomes 0.8201 and clears the four-fifths screen once its fifteen-person comparator grows; an inference about label bias drawn from a classifier with target leakage (AUC 0.9942 → 0.7915 once repaired); and an arithmetic claim that 0.8956 failed a 0.80 screen. We report the retractions because the retracted versions were the more quotable ones.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Social and professional topics** → *Governmental regulations*; • **Human-centered computing** → *Human computer interaction (HCI).*

KEYWORDS

algorithmic fairness, audit, disparate impact, intersectionality, lending, large language models, measurement

1 INTRODUCTION

A disparate impact ratio of 0.86 is not a finding. It becomes one when attached to who was denied credit, what recourse they had, and what the lender is obliged to check. Lee et al. [22] make this the organising commitment of human-centered fair AI: fairness is defined relative to a human decision context, not in metric space. We adopt that commitment and follow it one step further than is usual. If a fairness claim depends on its decision context, then so does the apparatus that produces the claim — and that apparatus can be wrong in ways that are invisible from inside it.

This is not a hypothetical. Deng et al. [10] find practitioners using fairness toolkits in ways their designers did not anticipate, and Lee and Singh [23] map systematic gaps in what open-source toolkits actually support. Holstein et al. [17] report that practitioners' needs diverge sharply from what the research literature supplies. Mei et al. [28] show, in a speech setting, that audit *pipelines* introduce pitfalls that change conclusions. Zaccour et al. [39] show that quantitative audits are more often blocked by data access than by method. The common thread is that the instrument is part of the finding.

We built a fairness audit across three lending decisions — consumer credit approval, mortgage origination, and peer-to-peer default risk — with a deterministic metric layer and a language model supplying the interpretive narrative. The intent was a benchmark. What the exercise produced, and what we report here, is a set of findings about audit infrastructure, three of which implicate our own instruments.

Claim. Fairness audit infrastructure infers normative categories from syntactic features — a label value, a character count, a row count, a threshold borrowed from another statute — and the inference fails hardest on the inputs where judgement was most needed. We support this with four demonstrations:

(1) **Metric orientation** (§4). Correct toolkit, clean data, sound metric, and a maximally wrong conclusion — disparate impact 0.0 where the true value is 0.9931 — produced by one convention about which label counts as favorable.
(2) **Reporting floors** (§5). An $n \geq 15$ floor suppressed five of eleven cells, three of them worse than anything the audit reported. The cost of a prudent statistical threshold falls entirely on the smallest groups.

(3) **The wrong statute** (§6). Auditing age at 40+, a threshold from employment law, returns a clean result on the one age class credit law actually protects.
(4) **Mechanical proxies for judgement** (§7). Three instruments we wrote in good faith systematically punished the correct response. We did not repair them, because repairing them would have hidden the finding.

We further report (§8) that the interpretive layer is contestable, and that its instability is model-generation-specific: prompt phrasing moves severity verdicts on `gpt-4o` by up to 14.7 points and by 0.0 on `gpt-5-mini`, on identical input.

What we retract. Three findings we previously published do not survive this re-analysis, and we set them out in place rather than quietly dropping them (§5, §5.1). Audit infrastructure that cannot be corrected in public is the problem this paper is about.

Position. We are not arguing that fairness metrics are useless or that audit tooling should be distrusted wholesale. We are arguing something narrower and more actionable: the failure modes documented here are not user error, and they are not exotic. Each arises from a defensible design decision made by someone who understood the domain. That is what makes them worth reporting at this venue.

2 RELATED WORK

Human-centered framing. Lee et al. [22] is our framing exemplar: a fairness claim is meaningful only relative to a human decision context. The line of work extending it has moved steadily toward stakeholder involvement in the choice of fairness criteria rather than in the interpretation of a fixed metric. Luo et al. [25] develop a protocol for explaining, negotiating and agreeing fairness metrics with stakeholders; Luo et al. [26] document how much disagreement stakeholders exhibit even when shown identical evidence; Luo et al. [24] extend this to affected individuals rather than institutional stakeholders. Deshpande and Sharp [11] ask who the stakeholders of a responsible AI system even are. Stumpf et al. [36] argue that fairness and correctness need user-centred assessment rather than metric reporting. Kim et al. [19] survey how equity is operationalised across HCI and fairness research and find the construct fragmenting under practical demand.

Our contribution is adjacent rather than competing. This literature asks who should choose the criteria. We ask what the apparatus does once a criterion has been chosen, and show that the apparatus can invert the answer without anyone choosing anything.

Audit practice and its instruments. Metaxa et al. [29] survey algorithm auditing as a method. Raji and Buolamwini [32] demonstrate that external audits change vendor behaviour, and Raji et al. [33] give an internal end-to-end audit framework. Kroll [21] argues for traceability as the operational principle behind accountability, which is the design stance our frozen-artifact approach takes. Documentation interventions — model cards [30], datasheets [15] — and process interventions such as co-designed checklists [27] share our premise that fairness work needs infrastructure, and share the risk we report: a checklist is a mechanical proxy for judgement, and §7 is a case study in what that costs.

Mei et al. [28] is the nearest prior work: audit pitfalls in an ASR setting that change the conclusion. Zamfirescu-Pereira et al. [40] diagnose human error inside an image-analysis pipeline. We add a lending case in which the pitfall is a metric convention and the affected quantity is the headline number.

Measurement. That fairness criteria conflict is settled [7, 20], and that the choice among them is not a technical matter is argued forcefully by Selbst et al. [35], whose abstraction traps describe our setting well. Corbett-Davies and Goel [9] catalogue how fairness measures mislead when detached from decision context. Barocas and Selbst [1] and Wachter et al. [37] establish that the legal notion of discrimination does not reduce to a metric threshold — which is why we treat the four-fifths screen as a screening signal throughout and make no legal claim. On intersectionality, Buolamwini and Gebru [5] provide the canonical empirical demonstration that marginal accuracy hides joint-group failure, and Foulds et al. [14] give a formal definition. §5.1 contributes a lending instance where the marginal audit passes cleanly and the joint audit does not.

Language models as auditors and judges. Using an LLM to interpret metrics invites the LLM-as-judge critique [41]: model-graded evaluation inherits the grader's biases. Track Q avoids the worst of this: its ground truth is a deterministic rule over computed metrics, never another model's opinion (§3). The one cross-model grade we report is flagged as secondary evidence (§8). Most directly, Wester et al. [38] study when and how LLMs deny user requests, which is the exact phenomenon our refusal detector was built to catch and, as §7 reports, systematically failed to measure.

3 SETTING AND METHOD

The benchmark is method here, not contribution; the artifact claim is made elsewhere. What matters for this paper is that every number an LLM saw was computed by deterministic code, and that the model was never permitted to recompute one.

3.1 Three lending decisions

The three use cases are deliberately not three instances of one setting (Table 1). They differ in the two properties that most change what an audit can conclude: whether the protected attribute is a protected class or an economic proxy, and whether the classifier discriminates enough for disparity to be measurable at all. Lending discrimination is documented empirically across this span — appearance and gender effects in peer-to-peer markets [12, 31], FinTech-era consumer lending [2], local credit-market conditions [6], and historical records used as training data [34].

3.2 Deterministic layer

For each protected attribute we compute disparate impact (DI), demographic parity difference (DPD), equal-opportunity difference (EOD), average odds difference (AOD) and the Theil index with AIF360 [3], cross-checked against Fairlearn [4]. A fixed rule maps metric values to a severity label (LOW, MODERATE, HIGH, CRITICAL); the CRITICAL boundary for DI is 0.72. Subgroup cells below $n = 15$ are reported with their n and excluded from conclusions. Data are German Credit [16], HMDA Georgia [8], and a Lending Club peer-to-peer default task.

Table 1: The three use cases. The last row governs how every downstream result reads.

	UC1 German Credit	UC2 HMDA (GA)	UC3 Lending Club
Decision	consumer credit	mortgage origination	P2P default risk
Attributes	sex, age, foreign worker	race, sex, age	gender, income, loan size
Type	protected class	protected class	1 class + 2 proxies
Favorable	good credit (1)	approved (1)	non-default (**0**)
Classifier	discriminating	discriminating	**~99.5% one class**

Two conventions are pinned by regression tests because both bit us: which label counts as favorable (§4), and the sign of AOD.

3.3 Two tracks, kept separate

The interpretive layer is evaluated along two axes that we never score against each other, because they answer different questions.

Track Q asks whether the model interprets fixed metrics reliably. For each of nine (use case × attribute) pairs we issue four prompts — `zero_shot`, `chain_of_thought`, `self_critique`, `constrained` — yielding 36 `gpt-4o` calls, each scored against the deterministic severity label. All 36 prompts are frozen and released, and a replay path reproduces the entire scoring pipeline with no API key.

Track H asks whether the narrative form raises the ceiling on audit quality. Three `o3` audits, one per use case, judged only against a five-criterion checklist, a guardrail baseline, and a quality rubric — never against the deterministic severity numbers, which would merely re-run Track Q.

Two detectors run over every response: a fabricated-citation check (n-gram match against the titles actually retrieved) and a fabricated-metric check (any numeric claim absent from the provided context). Spend is capped cumulatively at $15 with a pre-call abort and a persisted ledger. The ledger records **98 calls totalling $0.8933**; an additional $0.1125 was spent on a pre-freeze run whose calls predate the ledger. We state calls and dollars separately because an earlier version of this paper conflated them.

4 FINDING 1: METRIC ORIENTATION INVERTS THE HEADLINE NUMBER

In UC3, `gender` initially returned DI = 0.0 and a CRITICAL severity label — the most severe output the pipeline can produce, and on its face evidence of near-total exclusion.

It was an artifact. The prediction target is `loan_default`, so label 1 is the *unfavorable* outcome. Computing selection rates on predicted default gives 0.0000 for men and 0.0069 for women, and DI = 0.0000/0.0069 = 0.0. Oriented on the favorable outcome — predicted non-default — the rates are 1.0000 and 0.9931, so DI = 0.9931: near parity (Table 2).

Three things make this worth a section rather than a footnote.

Nothing was misused. AIF360 computes exactly what it is asked to. The data were clean. Disparate impact is the appropriate screen for this question. The error lives entirely in the mapping

Table 2: UC3 Lending Club, oriented on the favorable outcome (predicted non-default, label 0). `gender` previously read DI = 0.0, CRITICAL.

Attribute	DI	DPD	EOD	AOD
`gender` (priv. male)	**0.9931**	−0.0069	0.0	−0.0357
`income_level` (priv. med.)	1.0072	0.0072	0.0	0.0323
`loan_amount_level` (priv. med.)	1.0063	0.0062	0.0	0.0270

from a column name to a normative category, and that mapping is a convention rather than a computation. A target named `approved` makes label 1 favorable; a target named `loan_default` inverts it. Nothing in the metric signature records which situation holds.

The failure is maximally severe and silent. The reported value was wrong by a factor limited only by floating point, and it was wrong in the alarming direction, which is the direction least likely to prompt a second look: a CRITICAL disparity on a protected attribute reads as the audit working. Had the error run the other way it would have produced a false clean bill of health, which is the same defect with worse consequences.

It generalises. Default, delinquency, charge-off, fraud and churn are all conventionally coded with the adverse event as 1. Any fairness audit of a risk model — which is to say most fairness audits in lending — meets this condition. Deng et al. [10] and Lee and Singh [23] document toolkit-practice gaps of exactly this kind, and Selbst et al. [35] name the general form: the formalism omits the part of the problem that determines the answer.

Implication. Favorable-outcome orientation should be an explicit, required argument in fairness tooling, not inferred from label values or left to the analyst. It is now pinned by regression tests in our pipeline, which converts a silent inversion into a build failure. That is a small change with a large blast radius, and it is available to any toolkit maintainer today.

5 FINDING 2: THE REPORTING FLOOR ERASES THE LARGEST DISPARITIES

Subgroup reporting floors are standard practice and defensible: a ratio computed on six people is not evidence. Ours suppresses cells below $n = 15$. What that commitment does, in our data, is remove the most disadvantaged groups from the conclusion.

We hold the model fixed and vary only the audited population — a held-out test split, versus every row of the cleaned dataset scored out-of-fold by a model that never saw it. This isolates sample size from every other change (Table 3).

On the test split the audit returns MODERATE for `Black × Male` at 0.8169 while `American Indian × Female` sits at 0.1917, unreported. The audit is wrong about which group is most disadvantaged and wrong by two severity bands about how bad it is. Every suppressed cell in that run belongs to a racial minority.

At full population the same floor suppresses one cell of twelve, and the worst disparity — `American Indian × Male`, 0.6734, n = 31 — becomes visible. Nothing about the population changed; the audit simply stopped being run on a sample too small to see it.

The floor is not a bug, and we do not propose removing it. The point is narrower and harder to design around: a threshold chosen

Table 3: HMDA `race` × `sex` under an $n \geq 15$ floor. Identical leak-free model; only the audited population differs.

	Test split (n=2,196)	Full population (n=10,978)
Cells suppressed	**5 of 11**	1 of 12
Worst cell *reported*	0.8169 (MOD.)	0.6734 (CRIT.)
Worst cell that *exists*	**0.1917** (CRIT.)	0.6734 (CRIT.)
Suppressed cells worse than any reported	**3**	0

Table 4: UC1 German Credit marginals. All three clear the four-fifths screen.

Attribute	DI	DPD	EOD	AOD
`Sex` (priv. male)	0.8888	—	—	—
`AgeGroup` (priv. 40+)	0.9258	—	—	—
`foreign_worker` (priv. 0)	0.8387	—	—	—

for statistical prudence distributes its cost entirely onto the smallest groups, which in a lending context are the groups least able to absorb it. Two further details make it worse in practice. Our own codebase carries *two* floors, $n \geq 15$ in the drilldown and $n \geq 30$ in the qualitative layer, so the same cell is reportable or not depending on which module reads it. And the floor is applied silently: the audit output lists what passed, not what was withheld.

5.1 The marginal-versus-intersectional gap, corrected

Every marginal attribute in German Credit clears the four-fifths screen (Table 4). An audit reporting per-attribute disparate impact would return three passes and stop.

The intersection is the worst cell: `under_40 × female` has DI = 0.8201 (0.7012 against 0.8551 for `40_plus × female`), lower than any marginal attribute.

What we previously reported here, and why it was wrong. On the 200-row test split this cell read DI = 0.6087, which we published as CRITICAL. Its comparator was `40_plus × female` at $n = 15$ with a selection rate of exactly 1.0000 — fifteen people, all approved. A ratio against a saturated fifteen-person cell is not a severity grade. At full population that comparator holds 69 people at 0.8551 and the ratio rises to 0.8201, which *clears* the four-fifths screen.

The direction survives and the magnitude does not. We had also written that the disadvantaged group "is not identified by either single-attribute audit"; that was too strong, since `under_40 × male` at 0.7952 failed the screen on the same split, so the age effect was visible marginally. The honest claim is that marginal auditing understated a disparity it did point at.

This is the same defect as §5 seen from the other side: there, a small cell was suppressed and the disparity vanished; here, a small cell was retained as a *reference* and the disparity inflated. A floor governs which cells are reported. Nothing in the pipeline governs which cell becomes the comparator, and that choice moved a published verdict across two severity bands.

Table 5: UC2 HMDA (Georgia) marginals, leak-free model, full population (n=10,978). The banded `age_group` reads near parity while `age_62_plus` — the cut ECOA actually specifies — is the lowest of the four.

Attribute	DI	DPD	EOD	AOD
`race` (priv. White)	0.9371	—	—	—
`sex` (priv. Male)	0.9572	—	—	—
`age_group` (priv. mid, banded)	0.9791	—	—	—
`age_62_plus` (priv. under 62)	**0.8897**	—	—	—

5.2 UC2: the joint disparity exceeds both marginals

HMDA shows the same structure (Table 5). At full population `Black × Female` (n = 1,994) sits at DI = 0.8632, below both contributing marginals — the joint *disparity* exceeds the marginal ones, though the ratio is numerically lower, a phrasing we previously inverted.

Two caveats we did not previously state. First, the reference cell is chosen as the most-favoured adequately-sized subgroup, which is a winner's-curse estimator: measured instead against `White × Male`, the control group conventional in US fair lending analysis, the same cell reads 0.8388 on the test split and clears the screen. Reference-group choice alone moves the verdict across the four-fifths line, and nothing in the metric signature records which convention is in force. Second, and following §5, this cell is not the worst one; it is merely the worst *large* one.

5.3 Two readings a threshold gate destroys

The same tables carry two conclusions that survive only if the metrics are read together rather than screened individually.

A withdrawn inference. We previously read UC2's near-zero EOD and AOD alongside a low DI as evidence of label bias rather than differential model error. That reading is withdrawn. It rested on a classifier trained with `interest_rate` among its features, a column populated for 100.0% of originations and 0.7% of denials: the model was substantially detecting whether an interest rate existed, which is the outcome. Its errors were near zero in every group because it made almost no errors at all. Removing the leak drops AUC from 0.9942 to 0.7915. We report the retraction rather than the inference, because the inference was the more quotable of the two.

A non-finding is not a pass. UC1 `foreign_worker` has DI = 0.9402, comfortably clear. The privileged group has *six records*, against n = 194 on the other side and a reporting floor of $n \geq 15$. The ratio is dominated by the larger group and is not evidence of anything. Reported as a pass, it is worse than no result, because it closes an open question.

Implication. Intersectional analysis should be the default rather than an optional deepening, and per-attribute screening should be understood as the weakest available audit rather than the standard one. This is Buolamwini and Gebru [5]'s finding in a lending setting where the marginal audit does not merely understate the problem — it returns a clean result. As Foulds et al. [14] note, the number of subgroups grows combinatorially and statistical power falls with it, which is precisely why *n* must be reported beside every intersectional ratio rather than after it.

6 FINDING 3: THE AGE AUDIT USED THE WRONG STATUTE

Fairness work on credit routinely audits age at a 40+ threshold; our pipeline did. That number comes from the Age Discrimination in Employment Act, which governs employment. Credit is governed by the Equal Credit Opportunity Act, and Regulation B §1002.2(o) defines the protected class as **62 or older**. A comment in our own cleaning script asserted that 40+ "aligns with US ECOA". It does not.

The substitution is not cosmetic. Our banded `age_group` attribute pools young and senior applicants against the middle band. The two move in opposite directions — young applicants are approved at a higher rate than the middle band, senior applicants at a lower one — so pooling cancels them. The banded attribute returns DI = 0.9791, severity LOW: a clean bill of health. Audited on the statutory cut, the same model on the same rows returns **0.8897**, the largest age disparity in the dataset and the lowest of its four protected attributes (Table 5).

The flag needed to do this, `applicant_age_above_62`, was present and unused in the raw HMDA file. It is published precisely because §1002.2(o) turns on it, and it is necessary because the public age bands straddle 62 — the `55-64` band cannot be split, so no banded attribute can isolate the protected class. The information was there; the pipeline discarded it in favour of a threshold imported from the wrong body of law.

One further asymmetry this exposes and cannot resolve: §1002.6(b)(2) expressly permits *favouring* an elderly applicant. Disparate impact is symmetric, so a ratio above 1 on this attribute is lawful and is not a reverse-disparity finding, yet the metric reports it identically. The instrument cannot express the rule it is being used to check, which is the general form of this paper's claim stated in statute rather than in code.

7 FINDING 4: OUR OWN INSTRUMENTS PENALISED THE RIGHT ANSWER

The two findings above concern conventions we inherited. This one concerns code we wrote. Both instruments below were built deliberately, by authors who understood the domain, to hold the language model to account. Both systematically punished correct behaviour. We report them unrepaired.

7.1 The refusal detector measured brevity

Our refusal detector flagged six of nine responses under the `constrained` prompt strategy. *None was a refusal.* No flagged record contained a refusal phrase, a missing field, or an invalid severity value.

The detector treats any narrative field under 40 characters as effectively empty. The `constrained` strategy explicitly instructs the model to answer tersely. So `how_to_fix: "DisparateImpactRemover"` — a precise, correct, on-spec answer naming the applicable AIF360 post-processor in 22 characters — was recorded as a refusal. Compliance with the instruction was the trigger.

Reporting the raw output would have yielded a 66.7% refusal rate for `constrained` and a tidy narrative about constrained prompting inducing model evasion. That narrative would have been entirely an artifact of our measuring instrument. Wester et al. [38] study genuine LLM denials, and their work is the reason we built the detector; our detector does not measure the phenomenon they describe.

We did not change it. It is part of a frozen scoring harness, and altering it would invalidate the pre-freeze comparison in §8. Instead the analysis separates genuine refusals from brevity flags and reports both columns.

7.2 The guardrail punished "collect more data"

Our guardrail baseline requires every proposed mitigation to name a canonical algorithm — reweighing [18], disparate impact removal [13], calibrated equalised odds, and so on. It is a reasonable gate: it discriminates between a concrete remediation and vague advice, and the named methods are exactly the ones a practitioner would reach for.

Exactly two of nine attributes failed it. They are `foreign_worker_original` ($n = 6$ in the privileged group) and `loan_amount_level` (3.2× tier imbalance). In both cases the model proposed acquiring records to a stated floor, Bayesian hierarchical shrinkage, bootstrapped confidence intervals, and manual review until *n* is adequate — and in both cases that is the correct answer. No post-processor repairs an $n = 6$ subgroup. Applying one would manufacture false confidence in a quantity that cannot be estimated.

The gate encodes an assumption: that every fairness problem has an algorithmic mitigation. Where the problem is data scarcity, the assumption is false, and the gate penalises the only defensible response. The two failures are not noise — they are the two attributes where the assumption does not hold, which is the strongest form this finding could take.

We first suspected vocabulary matching and normalised the keyword list for hyphenation, since `o3` writes "Calibrated-Equalized-Odds" and "Kamiran-Calders Re-weighing". That fix was necessary and revealed the real signal rather than removing it: these two survive it. We did not widen the keyword list further, because doing so would convert a substantive finding into a passing score.

7.3 The fabrication detector penalises arithmetic

Our second detector flags any numeric value in a response that does not appear in the provided metric context, on the reasonable theory that an auditor should not invent numbers. Replaying the same packs against two models a generation apart shows what it actually measures.

`gpt-5-mini` trips 17 fabrication flags across 28 calls; `gpt-4o`, on identical input, trips 4. Read naively, the newer model fabricates five times as often. Checking each flagged value against the context, 14 **of** 21 **(**67%**) are derivable from it**: 88.88 is the `Sex` disparate impact of 0.8888 restated as a percentage, 83.87 is `foreign_worker`'s 0.8387, and most of the remainder are gaps between two context values expressed in percentage points. The detector matches raw float literals, so converting a ratio to a percentage or subtracting two provided numbers registers as fabrication.

Table 6: Track Q severity agreement. The ordering tracks classifier discriminativeness, not model capability.

Use case	Agreed	Rate
german_credit	4/12	33.3%
hmda	11/12	91.7%
lending_club	12/12	100.0%
Overall	**27/36**	**75.0%**

Table 7: Track Q by prompt strategy. `self_critique` leads on both axes; `zero_shot` carries five of eight hallucination flags.

Cycle	Strategy	Mean	Agreement	Halluc.
1	`zero_shot`	76.67	55.6%	5
2	`chain_of_thought`	80.00	77.8%	1
3	`self_critique`	**83.89**	**88.9%**	1
4	`constrained`	76.56	77.8%	1

`gpt-5-mini` does more of this arithmetic because it reasons more explicitly. The instrument therefore ranks the more capable model as the less trustworthy one, and it does so specifically because that model shows its work. Seven values (19.9 and 5.3, each repeated across cycles) were not derivable by this test and remain candidates for genuine fabrication; their repetition suggests a systematic derivation we have not identified rather than invention.

This is the same failure as §7.1, found independently in a second instrument written by the same authors on the same day: a mechanical proxy scoring a correct behaviour as a fault, in the direction that flatters the weaker system.

7.4 Why this belongs in the paper

Checklists, gates and detectors are the standard response to the problem that fairness work is hard to verify — the motivation behind model cards [30], datasheets [15] and co-designed checklists [27], and the operational content of internal audit frameworks [33]. Kroll [21] argues that traceability is what makes accountability actionable, and we agree.

The cost is that a mechanical proxy stands in for a judgement, and it will be wrong in the cases where judgement was most needed. Both of our instruments failed on the hardest inputs: the terse-but-correct answer, and the subgroup too small to analyse. Neither failure would have been visible from aggregate scores. Anyone reusing this harness inherits both proxies, which is why we document them rather than quietly fixing them.

8 THE INTERPRETIVE LAYER IS CONTESTABLE

Having argued that the deterministic apparatus carries commitments, we turn to the LLM layer built on top of it. The results support a narrow claim: the interpretive step is real work, it is unreliable in specific measurable ways, and its aggregate summary statistics should be distrusted.

8.1 Aggregate agreement is not a meaningful quantity

Overall severity agreement between `gpt-4o` and the deterministic labels is 27/36 = 75.0%. That number should not be quoted alone (Table 6).

The ordering is not a competence ranking. UC3's 100% is the easiest possible case: the classifier predicts non-default for roughly 99.5% of the test set (four default predictions in 600), so every attribute is a trivial near-parity LOW and agreeing costs nothing. UC1's 33.3% is the hardest: three MODERATE baselines sitting near threshold boundaries, where a small interpretive difference flips the label. Aggregate agreement is therefore a function of how discriminating the underlying classifiers are — a property of the benchmark, not of the model. A single headline agreement number for an LLM fairness auditor is not a meaningful quantity, and we would have reported one had we not decomposed it.

Errors are asymmetric in the safe direction: seven over-escalations against two under-escalations. Under-escalation is the consequential failure for an audit tool; over-escalation costs reviewer time.

8.2 Prompt phrasing moves the verdict

Three of nine attribute pairs changed severity label across the four prompt strategies on *identical* input context. `german_credit/Sex_original` went HIGH → HIGH → MODERATE → HIGH; `hmda/race` went LOW → MODERATE → MODERATE → MODERATE. Same model, same numbers, different wording, different answer. For an audit artifact that may be cited in a compliance conversation, this is the central reliability result.

8.3 Grounding suppresses fabrication

Comparing like-for-like against an earlier run over the same two use cases with the same model, where the only difference is whether retrieved literature was embedded in the prompt: fabricated-citation flags fall from 16 to 2, and mean score rises from 69.08 to 76.67. This is a design result rather than a model result, and it is the most actionable finding in the LLM half of the study: grounding the prompt in real retrieved evidence cuts fabrication roughly eightfold.

8.4 The provider dominates the prompt

Running the identical harness against `gemini-2.5-flash` produced refusals in 19 of 24 cycles, a mean score of 19.79, and 14 zero-score cycles. No prompt strategy compensates for that. Any benchmark of this shape measures provider capability at least as much as it measures method — a caveat that applies to our own numbers, and one reason we release the frozen prompts rather than only the scores.

8.5 What the narrative form added

Track H audits passed the citation check at 27/27 against the retrieved evidence pool, with weighted rubric scores of 4.8, 4.7 and 5.0 out of 5 (cross-model: `o3` generated, `gpt-4o` graded, reported as secondary evidence because a model grading a model is weaker than a mechanical gate [41]).

Two behaviours are worth naming because the deterministic template cannot produce them. On `foreign_worker_original` the audit independently reproduced the $n = 6$ caveat and quantified it: one label change would swing DI to 0.78, therefore statistical power is inadequate. On UC3 it identified the near-degenerate classifier unprompted and declined the flattering reading, converting a null result into a conditional prediction with a mechanism and a supporting citation — that near-parity here reflects a classifier approving almost everything, and that latent disparities may surface once realistic credit thresholds are enforced.

The deterministic pipeline establishes what is true. The narrative layer establishes what it means for a decision. They are not competing implementations, and asking which is more accurate is a category error.

9 DISCUSSION

Audit infrastructure needs its own audit. Our three findings share a structure: a component that is correct in its own terms produces a wrong or misleading conclusion at the level of the audit. Metric libraries, severity thresholds, refusal detectors and quality gates are all measurement instruments, and measurement instruments require validation against cases where the right answer is known independently. We had that only by accident — the $n = 6$ subgroup and the 22-character correct answer were adversarial test cases we did not design. Audit tooling should ship with them deliberately.

Conventions should become parameters. The orientation failure (§4) is repairable by design: make the favorable outcome a required argument. Generalising, any place where a fairness pipeline infers a normative category from a syntactic feature — a label value, a column name, a character count — is a place where the pipeline will silently encode someone's assumption. Making these explicit costs one parameter and converts a class of silent inversions into visible errors.

Which attributes deserve protection is deferred, not answered. UC3 audits `income_level` and `loan_amount_level` beside `gender` with identical machinery and identical thresholds. Near-parity on gender and near-parity on income level are not the same kind of result: one concerns a protected class, the other an economic characteristic that lending law generally permits pricing on. The pipeline cannot tell them apart, and by treating them uniformly it silently defers a normative question to whoever configured the attribute list. This is the point at which the stakeholder literature [11, 25, 26] becomes not merely complementary but necessary: the apparatus has no resources for the question.

Thresholds are screens, not verdicts. We report DI against the four-fifths screen throughout, and both §5.1 readings turn on refusing to treat it as a gate. A failed screen with EOD ≈ 0 means label bias; a passed screen at $n = 6$ means nothing at all. Barocas and Selbst [1] and Wachter et al. [37] establish why the legal question does not reduce to the ratio, and our results show the technical question does not either.

For practitioners. Three concrete moves follow: state the favorable outcome explicitly and test it; compute intersectional cells by default and print n beside every ratio; and treat each mechanical gate in the audit stack as a hypothesis to be falsified, starting with the inputs where the correct answer is unusual.

10 LIMITATIONS

Subgroup sizes. `foreign_worker_original` has $n = 6$ in the privileged group and several HMDA race categories are severely sparse. Intersectional cells are reported with n and excluded from conclusions below $n = 15$ — a floor whose cost we quantify in §5.

UC3 is barely informative as a fairness case. A classifier predicting one class $\sim$ 99.5% of the time cannot exhibit measurable disparity. UC3 exercises the pipeline; it does not audit a working model. Its value here is the orientation artifact and the proxy-versus-class contrast.

LLM nondeterminism. We pinned neither temperature nor seed. The four-cycle design varies prompt strategy, not repeated samples of one strategy, so cycle-to-cycle differences confound strategy with sampling noise. The severity-flip result in §8 should be read as "phrasing or sampling" rather than phrasing alone.

API versioning. `gpt-4o` and `o3` are moving targets. Frozen prompt packs make the inputs reproducible; the outputs are not.

Metric grounding. All metrics are associational. Root-cause mapping is rule-based inference over metric values and feature correlations, not causal identification. We make no causal claim.

Cost accounting. `o3` is priced at the higher of two conflicting internal rate tables, so reported spend is an upper bound rather than a billed amount.

No legal claim. ECOA, Regulation B and the four-fifths rule appear as decision context. A disparate impact ratio below 0.8 is a screening signal, not a finding of unlawful discrimination [1, 37].

Scope. HMDA is Georgia only; German Credit is 1990s German data used as a benchmark rather than a current population. Three datasets, one primary model, and a single institution's judgement about what counts as a correct mitigation.

A known artifact defect. Our consolidated metric CSV carries two disjoint column schemas because per-dataset files are concatenated without harmonisation. The per-dataset files are authoritative. We note it because it is exactly the class of infrastructure defect this paper is about.

11 CONCLUSION

We set out to benchmark whether a language model can interpret fairness metrics and found that the harder problem was upstream. A disparate impact ratio of 0.0 that should have been 0.9931; three passing marginals concealing an intersectional ratio of 0.6087; a refusal detector that measured brevity; a guardrail that penalised the only correct answer to a six-record subgroup. None of these was a bug in the ordinary sense, and none was user error. Each was a defensible design decision whose normative content became visible only when it punished a correct answer.

If fairness is defined relative to a human decision context [22], then the apparatus that measures fairness inherits that dependence. Our instruments were built by people who knew the domain and still encoded assumptions that inverted conclusions. We expect this to be the common case rather than the exception, and we release the frozen artifacts so that the claim can be checked against our own instruments rather than taken on our word.

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